

# fixed

## Fixed

**Revision:** 19 **Last modified:** 2026-06-01T17:11:08Z **Description:** Completed workable items (with evidence references) **Authority:** Constitution §11.4.93 (workable-items DB single source of truth) **Generated-by:** scripts/docs/db\_to\_md.py (DB is canonical; edit via cmd/hxc-registry, not by hand)

Total completed: **131**.

| HXC      | Type    | Pri | Title                                                                     | Commit  |
|----------|---------|-----|---------------------------------------------------------------------------|---------|
| HXC-001  | Task    | P1  |                                                                           |         |
| HXC-002  | Task    | P1  |                                                                           |         |
| HXC-003  | Task    | P1  |                                                                           |         |
| HXC-004  | Task    | P1  |                                                                           |         |
| HXC-005  | Task    | P1  |                                                                           |         |
| HXC-006  | Task    | P1  |                                                                           |         |
| HXC-007  | Task    | P1  |                                                                           |         |
| HXC-008  | Task    | P1  |                                                                           |         |
| HXC-009  | Task    | P1  |                                                                           |         |
| HXC-010  | Task    | P1  |                                                                           |         |
| HXC-011  | Task    | P1  |                                                                           |         |
| HXC-012  | Task    | P1  |                                                                           |         |
| HXC-013  | Task    | P1  |                                                                           |         |
| HXC-014  | Task    | P1  |                                                                           |         |
| HXC-015  | Task    | P1  |                                                                           |         |
| HXC-1001 | Task    | P0  | Stand up embedded-etcd integration harness for pkg/etcd tests             | dbed10c |
| HXC-1002 | Feature | P0  | Prove etcd Put/Get round-trip and missing-key semantics against real etcd | dbed10c |
| HXC-1003 | Feature | P1  | Prove etcd GetPrefix returns exactly matching multi-key set               | dbed10c |
| HXC-1004 | Feature | P1  | Prove etcd Delete/DeletePrefix remove keys (sink-side absence)            | dbed10c |
| HXC-1005 | Feature | P0  | Prove etcd Watch fires correct event and closes on ctx cancel             | dbed10c |

| HXC      | Type    | Pri | Title                                                                                             | Commit                  |
|----------|---------|-----|---------------------------------------------------------------------------------------------------|-------------------------|
| HXC-1006 | Feature | P1  | Prove etcd Lease/PutWithLease/RevokeLease/KeepAlive TTL behavior                                  | dbed10c                 |
| HXC-1007 | Feature | P0  | Prove etcd distributed Lock mutual exclusion (highest-risk path)                                  | dbed10c                 |
| HXC-1008 | Task    | P2  | Add mutation-paired etcd New() default-timeout injection test                                     | dbed10c                 |
| HXC-1009 | Task    | P2  | Prove etcd error-wrapping on dead endpoint                                                        | dbed10c                 |
| HXC-1010 | Feature | P0  | Replace pkg/infra map-echo orchestrator with real service Boot via testcontainers                 | 85ee85e85f221075dfc5f03 |
| HXC-1011 | Feature | P0  | Make pkg/infra Health probe real services and report false on failure                             | 85ee85e85f221075dfc5f03 |
| HXC-1012 | Feature | P1  | Make pkg/infra Logs capture real sink-side output honoring tail/since/timestamps                  | 85ee85e85f221075dfc5f03 |
| HXC-1013 | Feature | P1  | Make pkg/infra Scale create real replicas verified via runtime query                              | 85ee85e85f221075dfc5f03 |
| HXC-1014 | Feature | P1  | Make pkg/infra VMspawn/VMSSSH establish a real session end-to-end                                 | 44db9182bb5abe8f00ff83a |
| HXC-1015 | Feature | P2  | Make pkg/infra VMSimulateFailure/VMSimulatePartition assert observable network effect             | 85ee85e85f221075dfc5f03 |
| HXC-1016 | Feature | P0  | Add EtcdLocker integration tests against real etcd (block/acquire/lease)                          | dbed10c                 |
| HXC-1017 | Bug     | P0  | Rewrite TestMemoryLockerConcurrent to detect a broken lock without -race                          | dbed10c                 |
| HXC-1018 | Task    | P2  | Add explicit block-then-acquire ordering test for MemoryLocker                                    | dbed10c                 |
| HXC-1019 | Bug     | P2  | Characterize or fix MemoryLocker.Lock context-cancellation goroutine leak                         | 32feec3                 |
| HXC-1020 | Feature | P0  | Wire a real Builder behind pkg/build Service and prove image is pullable                          | 85ee85e85f221075dfc5f03 |
| HXC-1021 | Bug     | P1  | Drive pkg/build StateFailed via a genuine build failure not the 'fail' sentinel                   | dbed10c                 |
| HXC-1022 | Bug     | P1  | Make pkg/build List/Concurrent tests Start() the service and assert terminal states               | 32feec3                 |
| HXC-1023 | Task    | P2  | Replace time.Sleep polling in pkg/build with completion sync and add state-machine mutation tests | 32feec3                 |
| HXC-1024 | Bug     | P2  | Enforce or document pkg/build content-addressable cache digest integrity                          | dbed10c                 |
| HXC-1025 | Task    | P2  | Cover pkg/build cancel-mid-flight branch sets StateCancelled                                      | 32feec3                 |
| HXC-1026 | Feature | P0  | Add real mTLS handshake test for pkg/security (valid client succeeds, no/foreign cert rejected)   | dbed10c                 |
| HXC-1027 | Bug     | P1  |                                                                                                   | dbed10c                 |

| HXC      | Type    | Pri | Title                                                                                      | Commit                  |
|----------|---------|-----|--------------------------------------------------------------------------------------------|-------------------------|
|          |         |     | Assert MinVersion==TLS13 on the client<br>TLS path in pkg/security                         |                         |
| HXC-1028 | Feature | P0  | Locate/build production Vault client and<br>add dev-Vault integration test (KVv2 +<br>PKI) | 85ee85e85f221075dfc5f03 |
| HXC-1029 | Task    | P3  | Replace pkg/security non-nil-only<br>TestNewTLSConfigBuilder with<br>meaningful assertion  | 32feec3                 |
| HXC-1030 | Bug     | P1  | Verify pkg/hxcregistry item_history audit-<br>sink rows after Create/Update                | dbed10c                 |
| HXC-1031 | Bug     | P2  | Assert exact and stable<br>ComputeHeadingHash value in pkg/<br>hxcregistry                 | dbed10c                 |
| HXC-1032 | Task    | P2  | Add pkg/hxcregistry failure-path tests (dup<br>PK, missing-id Get/Update)                  | dbed10c                 |
| HXC-1033 | Bug     | P2  | Make pkg/hxcregistry concurrent test do<br>concurrent writes (or rename)                   | dbed10c                 |
| HXC-1034 | Task    | P3  | Prove pkg/hxcregistry timestamp round-trip<br>and DB CHECK-constraint enforcement          | dbed10c                 |
| HXC-1035 | Bug     | P0  | Wire pkg/wasm WasmPlugin.Execute to<br>real WASM output and prove transformed<br>result    | dbed10c                 |
| HXC-1036 | Task    | P1  | Add pkg/wasm negative tests for missing<br>execute export and corrupt module Init          | dbed10c                 |
| HXC-1037 | Bug     | P2  | Prove pkg/wasm init/shutdown exports<br>actually run via memory side-effect                | dbed10c                 |
| HXC-1038 | Task    | P3  | Cover pkg/wasm Host.Call<br>ErrUnexpectedResult branch                                     | 32feec3                 |
| HXC-1039 | Bug     | P1  | Replace self-defeating pkg/semaphore<br>TestSemaphore_ReleaseTooMany_Mutation              | a084eb1                 |
| HXC-1040 | Feature | P1  | Add pkg/semaphore M>>N concurrency<br>invariant test under -race                           | a084eb1                 |
| HXC-1041 | Task    | P3  | Add pkg/semaphore New(0) barrier and<br>exact-capacity boundary tests                      | a084eb1                 |
| HXC-1042 | Bug     | P1  | Convert pkg/storage error tests to errors.Is<br>on exported sentinels                      | dbed10c                 |
| HXC-1043 | Bug     | P2  | Reconcile and test pkg/storage FileStore vs<br>MemoryStore copy semantics                  | dbed10c                 |
| HXC-1044 | Task    | P2  | Add pkg/storage overwrite, missing-delete,<br>and List edge tests for both stores          | dbed10c                 |
| HXC-1045 | Feature | P1  | Add pkg/storage -race concurrent Put/Get/<br>Delete test to prove RWMutex                  | dbed10c                 |
| HXC-1046 | Bug     | P1  | Gate -race in CI for pkg/metrics<br>concurrency mutation tests                             | a084eb1                 |
| HXC-1047 | Bug     | P2  | Complete pkg/metrics Prometheus<br>exposition assertions (+Inf, _sum, ordering)            | a084eb1                 |

| HXC      | Type    | Pri | Title                                                                                       | Commit  |
|----------|---------|-----|---------------------------------------------------------------------------------------------|---------|
| HXC-1048 | Task    | P2  | Add pkg/metrics duplicate-registration and histogram edge tests                             | a084eb1 |
| HXC-1049 | Task    | P3  | Add pkg/metrics /metrics integration test via httptest.Server                               | a084eb1 |
| HXC-1050 | Bug     | P1  | Test pkg/grpccutil LoggingStreamInterceptor recv/send counters                              | a084eb1 |
| HXC-1051 | Bug     | P2  | Strengthen pkg/grpccutil log assertions to dynamic behavior-dependent fields                | a084eb1 |
| HXC-1052 | Task    | P2  | Cover pkg/grpccutil status.FromError code extraction for handler errors                     | a084eb1 |
| HXC-1053 | Bug     | P1  | Add pkg/context<br>TestDetach_PreservesValues for the value-retention guarantee             | a084eb1 |
| HXC-1054 | Bug     | P2  | Strengthen pkg/context WithTimeout deadline and expiry assertions                           | a084eb1 |
| HXC-1055 | Bug     | P1  | Give pkg/workerpool<br>TestPool_NilWorkSafe_Mutation a real sink                            | a084eb1 |
| HXC-1056 | Feature | P1  | Add pkg/workerpool fan-in count==N, post-Stop Submit, panic-recovery, idempotent-Stop tests | a084eb1 |
| HXC-1057 | Bug     | P1  | Make pkg/errors concurrency test honest via locked GetFields path + -race                   | a084eb1 |
| HXC-1058 | Task    | P2  | Pin pkg/errors Code enum exact values, distinctness, and WithFields nil receiver            | a084eb1 |
| HXC-1059 | Bug     | P2  | Replace tautological pkg/backoff cap-mutation test and pin Default()                        | a084eb1 |
| HXC-1060 | Bug     | P2  | Assert pkg/classads parser AST Op per operator and paren precedence                         | a084eb1 |
| HXC-1061 | Task    | P3  | Add pkg/classads eval failure-path and regexp non-match tests                               | a084eb1 |
| HXC-1062 | Bug     | P2  | Add pkg/crypto SHA-256 known-answer vectors and sentinel errors.Is checks                   | dbed10c |
| HXC-1063 | Task    | P3  | Add pkg/crypto AES-128/192, wrong-key decrypt, short-ciphertext, PBKDF2 vector tests        | dbed10c |
| HXC-1064 | Bug     | P1  | Add pkg/log prefix-field proof catching the SetLevel prefix-loss bug                        | a084eb1 |
| HXC-1065 | Task    | P3  | Strengthen pkg/log Warn test, add Debug-suppression, remove dead captureOutput scaffolding  | a084eb1 |
| HXC-1066 | Task    | P3  | Add pkg/lru boundary, cold-miss, and struct-value round-trip tests                          | a084eb1 |
| HXC-1067 | Feature | P2  | Add pkg/pubsub -race concurrency test and explicit drop-boundary assertion                  | a084eb1 |
| HXC-1068 | Bug     | P2  | Prove pkg/ratelimit bucket starts full and add Limiter conformance + -race tests            | a084eb1 |

| HXC      | Type    | Pri | Title                                                                             | Commit                  |
|----------|---------|-----|-----------------------------------------------------------------------------------|-------------------------|
| HXC-1069 | Bug     | P1  | Bring pkg/retry DoWithResult to behavioral parity with Do                         | a084eb1                 |
| HXC-1070 | Task    | P2  | Add pkg/retry transient-recovery, jitter-bound, and computeDelay table tests      | a084eb1                 |
| HXC-1071 | Task    | P3  | Strengthen pkg/serde MustMarshal exact-content and add Unmarshal failure tests    | a084eb1                 |
| HXC-1072 | Bug     | P2  | Pin pkg/validator error contract via errors.Is and remove/wire unused sentinels   | a084eb1                 |
| HXC-1073 | Task    | P3  | Add pkg/validator uint min/max, IsValidID empty, oneof/email boundary tests       | a084eb1                 |
| HXC-1074 | Task    | P0  | Establish fleet-wide -race CI gate across concurrency-sensitive packages          | a084eb1                 |
| HXC-1075 | Feature | P0  | Implement NATS/JetStream client wrapper (pkg/events)                              | 44db9182bb5abe8f00ff83a |
| HXC-1076 | Feature | P0  | Implement etcd client wrapper (pkg/etcd)                                          | 44db9182bb5abe8f00ff83a |
| HXC-1077 | Feature | P0  | Implement etcd discovery backend (pkg/discovery)                                  | 44db9182bb5abe8f00ff83a |
| HXC-1078 | Feature | P0  | Implement real Kafka producer/consumer (pkg/pubsub)                               | 44db9182bb5abe8f00ff83a |
| HXC-1079 | Feature | P0  | Implement cgroups v2 + /proc resource reader (pkg/resources)                      | 44db9182bb5abe8f00ff83a |
| HXC-1080 | Feature | P0  | Implement Omega-model scheduler core (pkg/scheduler, internal/scheduler)          | 44db9182bb5abe8f00ff83a |
| HXC-1081 | Feature | P0  | Implement optimistic-concurrency versioning for scheduler pool                    | 44db9182bb5abe8f00ff83a |
| HXC-1082 | Feature | P1  | Implement HTCondor ClassAds matching in scheduler (pkg/classads)                  | 44db9182bb5abe8f00ff83a |
| HXC-1083 | Feature | P1  | Implement gang scheduling and preemption in scheduler                             | 44db9182bb5abe8f00ff83a |
| HXC-1084 | Feature | P0  | Upgrade tmux backend to control-mode (-CC) with streaming I/O (pkg/session)       | 1367dca3bd532a491adb1a  |
| HXC-1085 | Feature | P0  | Implement CRDT session state and lifecycle (create/attach/migrate)                | 1367dca3bd532a491adb1a  |
| HXC-1086 | Feature | P0  | Implement real Node gRPC handlers serving helixv1 Node API (internal/node)        | 4d0ac6681bf586c30aaa881 |
| HXC-1087 | Feature | P0  | Make node registry etcd-backed and cross-process shared (internal/node)           | 2f27ac9001d3fc0a900f038 |
| HXC-1088 | Feature | P0  | Implement SWIM gossip membership and failure detection (pkg/swim)                 | 4d0ac6681bf586c30aaa881 |
| HXC-1089 | Feature | P0  | Implement WireGuard mesh config-gen and key rotation (pkg/wireguard)              | 2f27ac9001d3fc0a900f038 |
| HXC-1090 | Feature | P1  | Implement WireGuard NAT traversal / STUN hole-punch (pkg/wireguard)               | 1367dca3bd532a491adb1a  |
| HXC-1091 | Feature | P0  | Wire API gateway reverse proxy with prefix routing and /health (internal/gateway) | 2f27ac9001d3fc0a900f038 |

| HXC      | Type    | Pri | Title                                                                            | Commit                  |
|----------|---------|-----|----------------------------------------------------------------------------------|-------------------------|
| HXC-1092 | Feature | P0  | Add JWT/RBAC auth middleware to API gateway (internal/gateway)                   | 2f27ac9001d3fc0a900f038 |
| HXC-1093 | Feature | P0  | Implement leader election with fencing tokens (pkg/leader)                       | 4d0ac6681bf586c30aaa881 |
| HXC-1094 | Feature | P0  | Implement distributed lock primitives over etcd (pkg/lock)                       | 2f27ac9001d3fc0a900f038 |
| HXC-1095 | Feature | P1  | Implement Postgres-backed internal artifact registry (pkg/hxcregistry)           | 1367dca3bd532a491adb1a  |
| HXC-1099 | Feature | P1  | Implement Build Service orchestrator and worker pool (internal/build)            | 1367dca3bd532a491adb1a  |
| HXC-1101 | Feature | P0  | Implement e2ee transport record protocol (security/pkg/e2ee)                     | e12003abbb68db0e08fa13  |
| HXC-1102 | Feature | P1  | Implement software-rooted attestation (security/pkg/attestation)                 | e12003abbb68db0e08fa13  |
| HXC-1103 | Feature | P0  | Implement RBAC scopes and identity bindings (internal/security)                  | e12003abbb68db0e08fa13  |
| HXC-1104 | Feature | P1  | Implement SPIFFE/SPIRE SVID issuance and mTLS identity (internal/security)       | 7634fbbfaf4e66eba233c79 |
| HXC-1105 | Feature | P0  | Implement htmux CLI raw-mode terminal client (cmd/htmux)                         | e12003abbb68db0e08fa13  |
| HXC-1106 | Feature | P0  | Implement session I/O PTY-over-WebSocket forwarding (cmd/helix-session)          | e12003abbb68db0e08fa13  |
| HXC-1107 | Feature | P0  | Implement WebSocket envelope and MessagePack framing (pkg/websocket)             | e12003abbb68db0e08fa13  |
| HXC-1108 | Feature | P1  | Implement HelixQA live-service Challenge runner (cmd/helix-test)                 | 6c825dfc56a498b371578d  |
| HXC-1109 | Feature | P1  | Implement deterministic-sim, chaos, and device-sim test harness (cmd/helix-test) | 6c825dfc56a498b371578d  |
| HXC-1114 | Feature | P0  | Provision PostgreSQL primary schema with 15 tables, indexes, and triggers        | e12003abbb68db0e08fa13  |
| HXC-1115 | Task    | P0  | Define etcd key namespace and key-builder constants (pkg/etcd)                   | e12003abbb68db0e08fa13  |
| HXC-1116 | Feature | P1  | Define Redis cache, routing, and rate-limit key structure (pkg/storage)          | 7634fbbfaf4e66eba233c79 |
| HXC-1118 | Feature | P1  | Implement REST API surface on API Gateway with OpenAPI 3.0                       | 6c825dfc56a498b371578d  |
| HXC-1120 | Task    | P1  | Configure Kafka topics with partitions/replication/retention                     | 6c825dfc56a498b371578d  |
| HXC-1121 | Task    | P0  | Configure NATS JetStream streams (HELIX_NODES/SESSIONS/SCHEDULER/HEALTH/ALERTS)  | 6c825dfc56a498b371578d  |
| HXC-1127 | Feature | P1  | Implement OPA/WASM Policy Engine with HelixConstitution enforcement              | 7634fbbfaf4e66eba233c79 |
| HXC-1132 | Feature | P1  | Implement Metrics Collector node scrape endpoint and GPU metrics aggregation     | 7634fbbfaf4e66eba233c79 |

| HXC      | Type    | Pri | Title                                                               | Commit                 |
|----------|---------|-----|---------------------------------------------------------------------|------------------------|
| HXC-1133 | Feature | P1  | Implement token-bucket rate limiter for API gateway (pkg/ratelimit) | e12003abbb68db0e08fa13 |
| HXC-902  | Task    | P1  |                                                                     |                        |
| HXC-903  | Task    | P1  |                                                                     |                        |